NEAR EAST UNIVERSITY

Faculty of Artificial Intelligence and Informatics

Department of Software Engineering

INTERNSHIP REPORT

Summer Training

SWE299 & 399

Student: Ifediatu Mary Chisoanyi - 20226411

Supervisor: Mr Ofomata Chika .C.

Table of Contents

Introduction

Internships form a bridge between academic study and practical experience. For me, this internship with the ICT department of Alvan Ikoku Federal University in Nigeria was more than just a degree requirement. it was a chance to see how the theories I learned in the classroom apply to real systems and real users. The institution welcomed me into their team and involved me in daily operations, ongoing projects, and training activities that broadened my understanding of software engineering in practice.

Over the course of 40 days, I rotated through different responsibilities. In the beginning, I handled frontline technical support, helping staff and students with common issues such as software errors, printer faults, and password resets. I then participated in asset management tasks, updated the IT inventory, and gained basic exposure to system administration. Midway through, I was assigned two key projects: developing a software license inventory system and building a PC Diagnostics Checklist Tool. These assignments gave me the chance to design practical solutions that the department could use beyond my time there. I was also advised by my supervisor to complete a 10-day Coursera course on cybersecurity, which strengthened my ability to recognize threats and apply protective measures. Finally, I wrapped up my internship by creating three cybersecurity awareness posters aimed at students and staff, combining technical knowledge with simple, visual communication.

This report documents that journey. Each chapter covers a different stage of my internship: the tasks I performed, the tools I used, the challenges I encountered, and the lessons I carried away. The experience gave me not only technical skills but also insights into teamwork, communication, and the importance of preventive measures in ICT. What follows is a detailed account of my contributions and growth during this period of professional training.

Chapter 1: Technical Support and Troubleshooting

1.0 Overview

I spent the first weeks of my internship embedded with the ICT Service Desk of a Alvan Ikoku federal university in Nigeria. The team is small but busy. Each morning starts with the ticket queue, a quick stand-up, and then a steady flow of walk-ins from staff and students. My main job was to help people get back to work quickly, document what I did, and learn the department's standards along the way.

What follows is a detailed account of the kinds of issues I handled, the exact steps I took, and what I learned from each category of work.

1.1 Service Desk Workflow I Followed

I checked the shared mailbox and the ticketing board for new requests. For walk-ins, I created a ticket on the spot with the user's name, contact, device type, and a plain description of the issue in the user's own words. I also set a priority based on impact. A projector failure ten minutes before a lecture was urgent; a personal printer issue in a staff office could wait.

I used a repeatable checklist for each problem category. If I needed help, I asked a senior technician, took notes on what they tried, and added that to the knowledge base.

I never closed a ticket without a confirmation from the user that the fix worked. For recurring problems, I proposed a prevention step, like a driver update or a quick user tip posted to the notice board.

1.2.1 Software errors and application crashes

Typical cases included Office activation failures, missing DLL errors after updates, and apps that froze at startup. My approach:

1. Reproduce and observe. Ask the user to show the error. Capture the exact message.
2. Check basics. Storage space, OS updates pending, and whether the user has admin privileges when needed.

3. Logs and reliability monitor. On Windows, I checked Event Viewer and Reliability Monitor to see crash patterns. On Linux lab PCs, I looked at /var/log/syslog.
4. Repair or reinstall. For Office, I ran a quick repair first. If activation still failed, I checked network access to the licensing service and time/date settings, then retried activation.
5. Rollback or patch. For apps that broke after an update, I rolled back to the last stable version or applied the vendor's hotfix if available.

Example case. A staff laptop showed "Office activation required" even though it had been activated earlier. The date and time were off by several hours after a power drain. I corrected the system clock, connected to the campus network, ran Office repair, and activation completed. I documented the pattern and added "check date/time" to our activation checklist.

1.2.2 Printer problems

These were the most frequent walk-ins. I met three main categories: paper jams, "printer offline," and wrong driver or queue.

- Paper jams: I guided users to open the correct panel and remove jammed sheets without tearing them. I checked rollers for wear and made a note if the unit needed maintenance.
- Printer offline: I pinged the printer's IP from the user's PC. If that failed, I checked the network cable or Wi-Fi on the printer. If ping worked but jobs stuck in queue, I cleared the spooler and restarted the Print Spooler service.
- Wrong driver or queue: Many users picked a similar model from Windows' generic list. I removed the device, installed the exact driver from our repository, and re-added the printer on a TCP/IP port.

Example case. In a lab, half the PCs could not print. Their queues pointed to an old server share. I removed the shared printer, added the direct TCP/IP queue, disabled bidirectional support, printed a test page on each PC, and updated the lab image notes so this would be correct on future re-images.

1.2.3 Network and Wi-Fi access

I dealt with Wi-Fi login loops, "Connected but no internet," and devices that refused to join the secure SSID.

- Captive portal and certificates: I cleared saved networks, connected to the secure SSID, and made sure users entered full institutional emails. For devices stuck in loops, I checked browser settings and time sync. Certificate prompts were common on older Android phones; I guided users to accept the CA certificate when appropriate.

- IP conflicts: On desktops with static IPs, I verified the address was within the allowed range and not duplicated. I changed the IP or shifted the PC to DHCP where policy allowed.
- DNS issues: If web access failed by name but worked by IP, I flushed DNS and renewed the lease. I also checked that the DNS servers matched campus settings.

1.2.4 Password resets and account lockouts

Account issues came in waves, especially after students returned from break.

- I verified identity with student ID or staff card.
- I reset the password using the standard tool and guided the user to set a strong one.
- I reminded users to sign out and sign back in on all devices so the new password synced correctly.

Tip I shared with users. Update the password on your phone first. Old credentials cached in email apps can keep locking the account.

1.2.5 Slow or unresponsive computers

- Startup programs: I reviewed startup items and disabled non-essential entries.
- Disk health: I checked SMART status and free space. If a drive was near full, I moved large temporary files and recommended departmental storage.
- Malware scan: I ran a full scan with the approved antivirus and removed suspicious browser extensions.

1.3 Installing and Updating Software

I handled routine installs and updates for Windows and Linux.

For Windows, I used our offline repository of installers so I could work without relying on the internet. Where it made sense, I used batch scripts to install a set of tools in one go. On Linux, I used the package manager to install essential utilities for programming classes.

I scheduled updates during lunch hours or right after classes to avoid disrupting lectures. I learned to warn users about restarts and to save work before I began. If a restart was required, I stayed until the system came back up and tested the main apps.

I kept a simple sheet of which machines had which licensed software. This later helped when I worked on the license inventory project.

Example batch for a lab refresh.

- Remove outdated Java and install the current version.
- Install the latest PDF reader and Office updates.
- Verify the antivirus definitions are current and run a quick scan.
- Test with a sample document and a print job.

1.4 Documentation Habits I Built

Good notes saved me more than once.

- I wrote the symptoms, steps tried, and the final fix in plain language. If I learned a new trick from a senior tech, I credited them and captured the exact command or setting.
- When I solved something new, I turned my notes into a short how-to with a clear title. For example, "Fix for Wi-Fi login loop on Windows 10 after clean install."
- Where allowed, I captured settings before I changed them. This made it easy to roll back if a fix did not work.

Template I used.

- Problem summary
- Device and OS
- Exact error or behavior
- Steps taken (in order)
- Result and user confirmation
- Prevention or follow-up

1.5 Communication with Users

I learned to explain fixes without jargon. If I needed time to research, I said so and gave a realistic update time. For staff in a hurry, I offered a quick workaround first and then returned later to apply the permanent fix. I tried to leave users with one practical tip, like how to check printer status or how to spot a risky email.

1.6 What I Learned from Support Work

- Most problems have simple causes. Start with power, cables, network status, and time/date.
- Clear documentation turns a one-off fix into a repeatable process for the team.
- Users remember how you made them feel. Calm, polite help matters as much as technical skill.

Chapter 2: Inventory, Basic System Administration, and Cybersecurity Training

2.0 Overview

The second part of my internship focused on keeping track of the university's IT assets, learning the basics of system administration, and completing a 10-day cybersecurity awareness course. These three areas connected in practice. A complete asset list makes patching and licensing possible. Basic admin skills help you manage accounts safely. Cybersecurity training helps you see why these routines matter.

2.1.1 Why the inventory mattered

Without a current inventory, the ICT team cannot budget, plan replacements, or respond quickly to incidents. My tasks were designed to make the asset register more accurate and easier to use.

2.1.2 What data we tracked

For desktops, laptops, projectors, and a few network devices, I recorded:

- Device type and model
- Serial number and asset tag
- Assigned user or room
- Processor, RAM, and disk type
- Operating system and version
- Purchase year and warranty status
- Installed licensed software that needed tracking
- Physical condition notes

2.1.3 How I updated the register

The department used Excel to manage the asset register. I did three kinds of updates:

- I compared the existing list with what I found in rooms and offices. When a device had been moved, I updated the location and the responsible user.
- For newly purchased devices, I created full entries from the packing list and labels, then verified details on the device itself.
- I fixed typos, duplicated serials, and missing fields. I created drop-down lists for device types and locations to keep entries consistent.

2.1.4 Field visits and condition checks

I visited lecture halls and labs to tag equipment and check condition. On projectors, I noted lamp hours and evidence of overheating. On desktops, I checked front USB ports, optical drives where present, and fan noise that could indicate dust buildup. If I found anything that might fail soon, I opened a maintenance ticket and attached my notes so a technician could schedule a visit.

2.1.5 Simple reporting and quality control

I built two pivot tables in Excel:

- Assets by department and device type
- Laptops by warranty status

These quick views helped answer common questions without searching the entire sheet. Before I marked a day's work as done, I sorted by serial number to spot duplicates and filtered for blank fields to fill in missing data.

2.1.6 Small automation that helped

Where policy allowed, I captured system info directly from Windows using built-in tools. This made entries more accurate.

2.2.1 User accounts and permissions

I learned how accounts are created for staff and students, and how access is granted to shared resources. My tasks included:

- Checking a user's group memberships when they could not access a shared folder
- Resetting passwords when authorized
- Removing stale accounts from groups when a user changed departments

I practiced safe habits: verify identity first, change only what is needed, and record the change in the ticket.

2.2.2 Understanding the server side

I spent time observing how the team used directory services to manage users and devices. I saw how organizational units separated students from staff, and how group policies applied settings like password rules and desktop restrictions in labs. I did not make changes on production systems, but I practiced on a non-critical workstation so I could learn without risk.

2.2.3 Everyday admin commands I used

On Windows workstations, I got comfortable with:

- ipconfig /all to check network configuration
- ping and tracer to test connectivity
- Event Viewer to read application and system logs

On Linux lab machines, I practiced:

- sudo apt update && sudo apt upgrade for updates
- Checking services with system status
- Basic file permissions with chmod and chown

2.2.4 Remote assistance

When a user was in another building, I used the department's approved remote support tool to see the screen, request control with permission, and guide them through steps. I always asked the user to close any personal documents first and explained exactly what I was doing.

Chapter 3: Software License Inventory

3.0 Introduction

During my internship, one of the significant projects I worked on was building a Software License Inventory for the ICT department. This task may not sound as hands-on as fixing a printer or troubleshooting a network error, but it is just as important. Without proper license tracking, organizations face serious risks: expired software that stops working mid-semester, unpatched applications that open security holes, or even legal issues if auditors discover unlicensed installations. In short, license management is the quiet backbone of ICT compliance and stability.

3.1 Objective of the Project

The goals I set for the license inventory project were:

- Create a single source of truth for all licensed software across the university.
- Track expiry dates so ICT staff could act before disruptions occurred.
- Map each license to a specific machine or user for accountability.
- Present the status of licenses visually, so at a glance staff could see what needed attention.

These goals shaped the structure of the spreadsheet and the rules I applied within it.

3.2 Designing the Inventory Fields

The spreadsheet was structured with clear, well-defined columns. Each entry captured the most critical details:

- Software Name – e.g., Windows 11 Pro, Microsoft Office 365, Adobe Photoshop.
- License Key – the alphanumeric activation key tied to the purchase.
- Expiry Date – the exact date the license would no longer be valid.
- Assigned Machine/User – for example, “Lab PC-05” or “Admin Laptop-02.”

- Status – a calculated column that displayed whether the license was *Up to Date, Due Soon,* or *Expired.*

I chose to include the status column because scanning through raw dates is inefficient. By color-coding the results, ICT staff can immediately see where to focus.

3.3 Populating the Inventory

To test and demonstrate the system, I entered over 35 software records that reflected the kinds of tools a university typically uses. I grouped them into categories:

- Operating Systems: Windows 10 Pro, Windows 11 Pro, Windows Server 2019
- Office Suites: Microsoft Office 2016, Office 2019, Office 365
- Antivirus & Security: Kaspersky, Norton, McAfee, ESET Endpoint Security
- Design & Media: Adobe Photoshop, CorelDRAW, AutoCAD
- Data & Analytics: SPSS, MATLAB, Stata, RStudio
- Specialized Academic Tools: EndNote, NVivo, Minitab
- Utilities: WinRAR, VMware Workstation, TeamViewer

Each entry included a unique key, a realistic expiry date spread across the next three years, and a machine or user assignment. This ensured that the inventory simulated an authentic environment, not just a generic list.

3.4 Conditional Formatting Rules

One of the most practical features was the visual status indicators I built using Excel's conditional formatting:

- Green (Up to Date): License valid for more than 30 days.
- Yellow (Due Soon): License expiring within 30 days.
- Red (Expired): License already past its expiry date.

With this setup, anyone opening the sheet could instantly identify urgent cases without scanning dozens of rows.

Example scenario. In my test data, a Kaspersky license assigned to "Lab PC-12" was set to expire in two weeks. The cell turned yellow, signaling it needed renewal soon. This simple alert system mimicked what the ICT team would need in real life to prevent downtime.

3.5 Benefits Observed

From building and testing the inventory, I identified several clear advantages:

- Improved Visibility: All license data is now in one place, easy to update.
- Compliance Assurance: By knowing which licenses are expiring, the university reduces legal and security risks.
- Budget Planning: Renewal costs can be anticipated and scheduled, avoiding last-minute requests.
- Accountability: Tying each license to a machine/user prevents duplication or “lost” licenses.
- Ease of Use: Because it is a spreadsheet, no extra training or special tools are required.

3.6 Limitations of the System

While useful, the inventory has some limitations:

- Manual entry is required, which means human error or delays are possible.
- Duplicate entries can occur unless checked carefully.
- It does not integrate directly with online license servers or dashboards.

Despite these issues, the benefits outweighed the drawbacks, especially since the solution is cost-free and easy to adopt immediately.

3.7 Future Improvements

During the project, I noted several potential upgrades:

- Automated reminders: Setting up email alerts when a license nears expiry.
- Database version: Using a database like SQLite or MySQL for better scalability and multi-user access.
- Web dashboard: A simple web interface where ICT staff could log in and update licenses, making the system more collaborative.

These improvements would transform the spreadsheet into a more advanced tool, but the current version already addresses the department’s most urgent needs.

3.8 Reflection and Conclusion

Building the software license inventory taught me that ICT management is not only about fixing broken devices. it is also about planning and prevention. I experienced how much difference a well-structured sheet can make in reducing risks. While I worked on other more visible tasks during my internship, this project showed me the value of quiet, behind the scenes work that ensures the entire system runs smoothly.

For the university, the inventory provides a simple but effective way to stay compliant and organized. For me as an intern, it was a chance to connect what I was learning in class about database structures and information systems with a real-world problem, solved in a way that is practical, low-cost, and immediately useful.

Chapter 4: PC Diagnostics Checklist Tool – Design and Development

4.0 Introduction

As part of my internship, I was tasked with building a PC Diagnostics Checklist Tool. The idea came from the ICT team's need for a consistent way to document the health of computers during routine checks. Until then, staff wrote notes on paper or saved them in scattered files, which made it hard to track a machine's history. My tool provided a lightweight desktop app that allowed staff to follow a checklist, add notes, and store everything in one log file. This project became one of the most practical contributions I made during the internship.

4.1 Purpose of the Tool

The main goal was to create a simple, consistent, and easy-to-use tool that could:

- Guide ICT staff through a standard checklist when diagnosing hardware and software.
- Provide a space for extra notes in case specific issues were found.
- Store all results in a single log file with timestamps, so every check became part of a running history.

By doing this, the tool improved accountability, reduced missed steps, and made it easier to review the condition of a computer over time.

4.2 Technology Used

To keep the tool accessible and lightweight, I built it using:

- Python: as the core programming language.
- Tkinter: Python's built-in GUI library, which avoided the need for external frameworks.
- Text file storage: All logs were saved into diagnostic_log.txt, ensuring portability and no extra setup.

These choices ensured the tool could run on most systems without special installation. Later, I modified it to always save logs in a fixed folder (Documents/DiagnosticsLogs) so users never had to search for the file.

4.3 Main Features

The app had three major components:

4.3.1 Checklist

Staff could go through a list of common checks and mark each item as either OK or Issue. The list included:

- Power supply
- Monitor
- Keyboard & Mouse
- Network connectivity
- Antivirus status
- Disk health
- Installed software updates

This ensured that no essential check was overlooked.

4.3.2 Notes Section

Alongside the checklist, I included a text box where staff could write additional comments. For example, if a PC failed the network check, they could note the error observed or steps taken to address it.

4.3.3 Save Log Button

When the user clicked Save Log, the app:

1. Gathered all the checklist responses and notes.
2. Added a timestamp.
3. Saved the entry into diagnostic_log.txt (or created the file if it didn't already exist).
4. Displayed a confirmation popup to reassure the user the log was recorded.

4.3.4 View Logs Button

Staff could press View Logs to open a scrollable window showing all past entries. This allowed them to quickly review a machine's history without leaving the app.

4.4 File Handling

The file handling system was deliberately simple:

- If diagnostic_log.txt didn't exist, the app created it automatically on the first save.
- All new logs were appended, keeping older entries intact.
- By default, logs were saved in the same folder as the app. After feedback, I modified the code so logs always saved in Documents/DiagnosticsLogs, making it easier to find and back up the data.

This ensured that staff could trust the logs would always be in one predictable place.

4.5 Typical Workflow for ICT Staff

The intended workflow was straightforward:

1. Open the app.
2. Go through the checklist, ticking items as verified.
3. Enter additional notes about any issues or observations.
4. Press Save Log (a confirmation popup appeared).
5. If needed, press View Logs to review past checks.

This structure encouraged consistency and reduced the chances of skipping a critical step.

4.6 Reflection on Development

Building the app taught me several lessons:

- Simplicity matters. The ICT staff appreciated that the tool ran without extra installation and saved logs in plain text.

- User feedback improves design. The fixed save directory was a direct result of staff feedback after testing.
- Logs become history. Over time, the log file grew into a valuable record of patterns e.g., repeated network issues in a lab could be identified and escalated.

Overall, this project helped me connect my programming knowledge with a real need in ICT operations. It was rewarding to see a tool I built being used daily by staff during maintenance checks.

Chapter 5: PC Diagnostics Checklist Tool – Testing and Practical Use

5.0 Introduction

After building the PC Diagnostics Checklist Tool, the next step was testing it in real ICT workflows. I wanted to ensure the app was not only functional but also practical and reliable for staff who might not be very technical. In this chapter, I describe how I tested the tool, how staff interacted with it, and what I learned from real usage.

5.1 Testing Methodology

I carried out both self-testing and user testing:

- Self-testing: I ran the tool on several lab and office PCs, entering different checklist results and notes. I deliberately tested edge cases, such as leaving the notes empty or marking all items as issues.
- User testing: I asked ICT staff to use the tool during their normal maintenance routines. I observed how easily they navigated the interface and whether they understood the flow without much explanation.

5.2 Observations During Testing

5.2.1 Functionality

- The checklist worked as intended. Staff liked having a structured list rather than relying on memory.
- The notes section was useful for adding context, such as "Network issue traced to faulty cable."
- The Save Log button always appended correctly to the log file, with timestamps in the right format.
- The View Logs feature was especially appreciated. One staff member commented that it felt like having a mini "medical record" for each PC.

5.2.2 File Management

- The decision to fix the log file location proved valuable. Staff no longer had to ask, “Where is the log saved?”
- Backups were easier because the entire DiagnosticsLogs folder could be copied at once.

5.2.3 Usability

- The interface was simple enough that staff used it with little guidance.
- The confirmation popup after saving gave users confidence that their work was not lost.
- The scrollable log window handled large entries without freezing.

5.3 Sample Test Case

Scenario: Running a check on Lab PC-07.

- Checklist results: Power supply OK, Monitor OK, Keyboard Issue (keys sticking), Network OK, Antivirus OK, Disk OK.
- Notes: “Keyboard replacement recommended; tested with spare keyboard to confirm.”
- Log saved with timestamp.

Later, viewing logs showed this entry appended after previous ones, creating a continuous history of Lab PC-07’s status.

5.4 Benefits Realized in Practice

- Consistency: Each PC was checked against the same list, reducing oversight.
- Documentation: Logs created a trail that could be referenced later for audits or recurring issues.
- Efficiency: Instead of rewriting the same problems in different formats, staff used the structured tool.
- Scalability: The log file grew naturally over time without becoming unwieldy.

5.5 Limitations and Areas to Improve

- The checklist was static. Adding or removing items required editing the code. A future version could load checklist items from an external file.
- The log file was plain text, which made searching or filtering harder compared to a database.
- The tool was designed for individual PCs, not for centralized multi-user access.

Chapter 6: Cybersecurity Course – Introduction to Cybersecurity Essentials

6.0 Introduction

One of the highlights of my internship was the opportunity to complete a 10-day online course titled “Introduction to Cybersecurity Essentials” on Coursera. My supervisor in the ICT department specifically advised me to take the course as part of my internship training, explaining that it would strengthen my understanding of information security and align with the department’s focus on safe ICT practices. By the end of the 10 days, I earned a certificate of completion, which served as both a personal achievement and evidence that I had built essential cybersecurity skills.

The course was divided into practical modules that combined theory with case studies and exercises. For me as an intern, it connected the routine ICT tasks I was already handling with the broader discipline of information security. Instead of only reacting to issues like malware infections or account lockouts, I started to see how these problems fit into the bigger picture of cybersecurity.

6.1 Understanding Common Threats

The first major focus of the course was learning how to identify common cybersecurity threats. We studied the most prevalent risks facing individuals and organizations:

- Malware: Viruses, worms, trojans, and ransomware, with real examples of how attackers trick users into installing them.
- Social engineering: Phishing emails, fake login portals, and impersonation tactics designed to manipulate human behavior.
- Data breaches: How weak access controls or misconfigured systems lead to unauthorized exposure of sensitive information.
- Other attack vectors: Denial-of-service (DoS) and man-in-the-middle attacks were introduced, showing how attackers exploit networks.

The course included interactive scenarios. For example, I had to identify whether an email was legitimate or a phishing attempt by looking at sender addresses, link destinations, and language cues. This directly reinforced what I saw during my internship when users brought in suspicious emails.

6.2 Securing Systems with Updates and Patches

Another key part of the training focused on software updates and patching. The course stressed that outdated systems are one of the easiest entry points for attackers. I learned that:

- Updates close known vulnerabilities before attackers can exploit them.
- Delays in patching increase risk, especially for widely used software.
- Patches should be applied on a schedule, but also tested to avoid disrupting workflows.

We also covered device hardening techniques such as disabling unnecessary ports, removing unused applications, and configuring firewalls properly. These practices directly related to my internship tasks where I updated software in labs and checked system logs after updates.

6.3 Authentication, Encryption, and Physical Security

The course emphasized that cybersecurity is not only about digital tools but also about layered defenses.

- Authentication: We studied the use of strong passwords, multi-factor authentication (MFA), and biometric systems. I learned how MFA reduces the impact of stolen passwords by requiring a second proof of identity.
- Encryption: I learned the basics of symmetric vs. asymmetric encryption and how encryption protects data at rest and in transit. A practical example was encrypting communication between a web browser and a server (HTTPS).
- Physical security: The course discussed measures like secure server rooms, access badges, CCTV, and environmental safeguards (fire suppression, temperature controls). These ensure that data centers remain operational even during physical incidents.

This module reminded me that in a university setting, physical access is sometimes overlooked. A locked office or secured lab is just as important as having a strong password.

6.4 Cybersecurity Best Practices

The course also introduced best practices that individuals and organizations should follow consistently:

- Password creation and management: Use long, unique passwords, avoid dictionary words, and enable password managers when possible.
- Use of MFA: Always enable multi-factor authentication where available.

- Regular updates: Keep operating systems, antivirus software, and applications current.
- Backups: Maintain secure and frequent backups to minimize damage in case of ransomware.
- User awareness: Conduct ongoing training to help users recognize phishing and other scams.

As part of the module, I practiced writing guidelines for staff and students. For example: "Use at least 12 characters, mix letters, numbers, and symbols, and never reuse the same password on different platforms."

6.5 Applying Knowledge in the Internship Context

While completing the course, I was able to apply what I learned to my day-to-day internship tasks:

- When handling password resets, I encouraged users to create stronger passwords and avoid obvious choices like names or birthdates.
- During lab updates, I double-checked that critical patches were installed and documented which machines were updated.
- When users reported suspicious attachments, I applied the phishing detection techniques from the course before escalating the cases.
- I also realized the importance of environmental safeguards when I saw projectors overheating in lecture halls, good ventilation and temperature control are also part of keeping ICT equipment safe.

6.6 Reflection and Conclusion

Completing the "Introduction to Cybersecurity Essentials" course gave me a clearer understanding of how day-to-day ICT support ties into the larger framework of cybersecurity. Instead of seeing tasks like updating antivirus or resetting passwords as routine chores, I began to view them as frontline defenses in a bigger system of protection.

I am grateful that my supervisor encouraged me to take this course. It felt like the institution was investing in me, ensuring that I didn't just learn technical troubleshooting but also gained a foundation in protecting systems and people from cyber risks. The certificate at the end of the course was not just a credential but evidence of a learning journey. It made me more confident in explaining threats to staff and students and in applying best practices in my own work. Most importantly, it showed me that cybersecurity is not an isolated specialty, it is a responsibility that cuts across every role in ICT.

Chapter 7: Cybersecurity Awareness Posters

7.0 Introduction

As part of my internship, I was encouraged to take what I had learned from the cybersecurity course and translate it into something that could directly reach staff and students. My supervisor suggested designing cybersecurity awareness posters that could be displayed in labs, offices, and hallways. The idea was to use short, clear messages with visuals so that even people who are not very technical could quickly grasp the importance of safe digital habits.

I designed three posters in Canva, each focusing on a different theme: Phishing Awareness, Password Safety, and Reporting Issues. The designs followed a consistent style, bold headlines, simple color schemes, and clean visuals. The goal was to make them both eye-catching and informative.

7.1 Poster 1: Phishing Awareness

Main Message / Slogan:
"Don't Take the Bait – Spot Phishing Emails!"

Supporting Points:

- Check the sender's email address.
- Don't click suspicious links or attachments.
- Verify requests for money or login details.

Visual Concept:
I used the idea of a fish hook catching an email icon, paired with an envelope marked with a red warning sign. The bold text "Think Before You Click" stood out in large font. The intention was to connect the term *phishing* with a literal hook so the message became instantly memorable.

7.2 Poster 2: Password Safety

Main Message / Slogan:
"Strong Passwords = Strong Protection"

Supporting Points:

- Use at least 8–12 characters.
- Mix letters, numbers, and symbols.
- Don't reuse the same password everywhere.
- Enable 2FA when possible.

Visual Concept:
This poster featured a cartoon padlock next to a password field showing a green checkmark for "strong." To emphasize the danger of weak passwords, I placed "123456" crossed out in red and replaced it with a stronger alternative. The lock-and-shield design reinforced the sense of safety.

7.3 Poster 3: Reporting Issues

Main Message / Slogan:
"See Something Suspicious? Report It!"

Supporting Points:

- Don't ignore security warnings.
- Report strange emails, pop-ups, or system errors.
- Contact the ICT Department immediately.

Visual Concept:
I used an exclamation mark symbol beside a computer screen with a red alert sign. Another element was an email icon with a highlighted "Report" button. To add a human touch, I included the silhouette of a person raising a hand, suggesting that speaking up helps protect the entire community.

7.4 Design Process

I created all three posters using Canva, choosing A4 size templates so they could be printed easily. I kept the designs consistent by using two main colors (blue and white) with red highlights to draw attention to warnings. The text was intentionally short and bulleted to ensure quick readability. Icons such as locks, email envelopes, and warning symbols were used instead of lengthy explanations.

To ensure the posters were effective, I asked colleagues in the ICT office for feedback. They appreciated that the visuals made the key ideas easy to remember. The posters were then shared digitally and printed for display in student labs.

7.5 Reflection and Conclusion

Designing these posters was different from troubleshooting or coding. It required me to think like a communicator, not just a technician. I learned that raising cybersecurity awareness depends on how clearly and simply information is delivered. While a long training session might overwhelm students, a well-placed poster with a bold message can catch their attention in seconds.

This task also showed me the importance of visual learning tools in ICT. By summarizing threats into simple slogans and graphics, I could help non-technical users build safer habits without needing to understand the technical details.

Overall, the posters became a bridge between the technical knowledge I gained from the cybersecurity course and the everyday actions needed from staff and students to keep the university secure.

Conclusion

Looking back on my internship, I can see how much I learned and how far I developed in a short span of 40 days. Every task from fixing small technical issues to designing an inventory system taught me something about the daily realities of ICT work. I learned that technical skill alone is not enough; patience, clear communication, and attention to detail are just as important when helping users or maintaining systems.

The projects I completed gave me confidence in applying my programming knowledge to solve real problems. The Software License Inventory and PC Diagnostics Checklist Tool are small but meaningful contributions, showing me how even simple tools can make a department more organized and effective. The cybersecurity course and poster designs reminded me that ICT is not only about machines but also about people how they use technology and how they can be guided to use it safely.

Most importantly, this internship confirmed my decision to pursue software engineering as a career. It showed me how my academic training fits into a professional setting and how I can keep growing with practice and learning. I am grateful to my supervisor and the ICT department for their support and guidance, and I believe the knowledge and experience I gained here will continue to influence my work and studies in the years ahead.

REFERENCE

1) U.S. National Science Foundation, *IT Inventory Standard*, University System of New Hampshire (USNH), 2022. Available: https://www.usnh.edu/it/departments/cybersecurity/cybersecurity-policies-standards/it-inventory-standard

2) MoldStud, *Managing Hardware and Software Inventory in University Systems: Tips for Administrators*, 2021. Available: https://moldstud.com/articles/p-managing-hardware-and-software-inventory-in-university-systems-tips-for-administrators

3) MoldStud, *Effective Strategies for Managing Software Licenses in University Systems*, 2021. Available: https://moldstud.com/articles/p-effective-strategies-for-managing-software-licenses-in-university-systems

4) Tkinter.com, *Save and Open To-Do Lists – Python Tkinter GUI Tutorial*, 2020. Available: https://tkinter.com/save-and-open-todo-lists-python-tkinter-gui-tutorial-156

5) Tkinter.com, *Read and Write to Text Files – Python Tkinter GUI Tutorial*, 2020. Available: https://tkinter.com/read-and-write-to-text-files-python-tkinter-gui-tutorial-100

6) GitHub Repository, *CheckList: A Checklist App with Tkinter (Save/Load Features)*, 2022. Available: https://github.com/GeorgeT59/CheckList

7) Coursera, *Introduction to Cybersecurity Essentials – Cisco Networking Academy*, Coursera, 2023. Available: https://www.coursera.org/learn/introduction-to-cybersecurity

8) ArXiv, *Want to Raise Cybersecurity Awareness? Start with Future IT Professionals*, arXiv:2307.07608, July 2023. Available: https://arxiv.org/abs/2307.07608

9) ArXiv, *Adopting Cybersecurity Concepts into Curriculum*, arXiv:2209.10407, Sept. 2022. Available: https://arxiv.org/abs/2209.10407

10) University of California, *Cybersecurity Posters – Security Awareness Resources*, UCOP, 2021. Available: https://security.ucop.edu/resources/security-awareness/cybersecurity-posters.html

11) SquaringTheNet, *Boost Cybersecurity with Engaging Awareness Posters: Tips and Strategies*, 2022. Available: https://squaringthenet.org/cybersecurity-awareness/boost-cybersecurity-with-engaging-awareness-posters-tips-and-strategies

12) ACM Digital Library, *Properties for Cybersecurity Awareness Posters' Design and Quality Assessment*, ACM, 2022. Available: https://dl.acm.org/doi/10.1145/3538969.3543794

APPENDIX

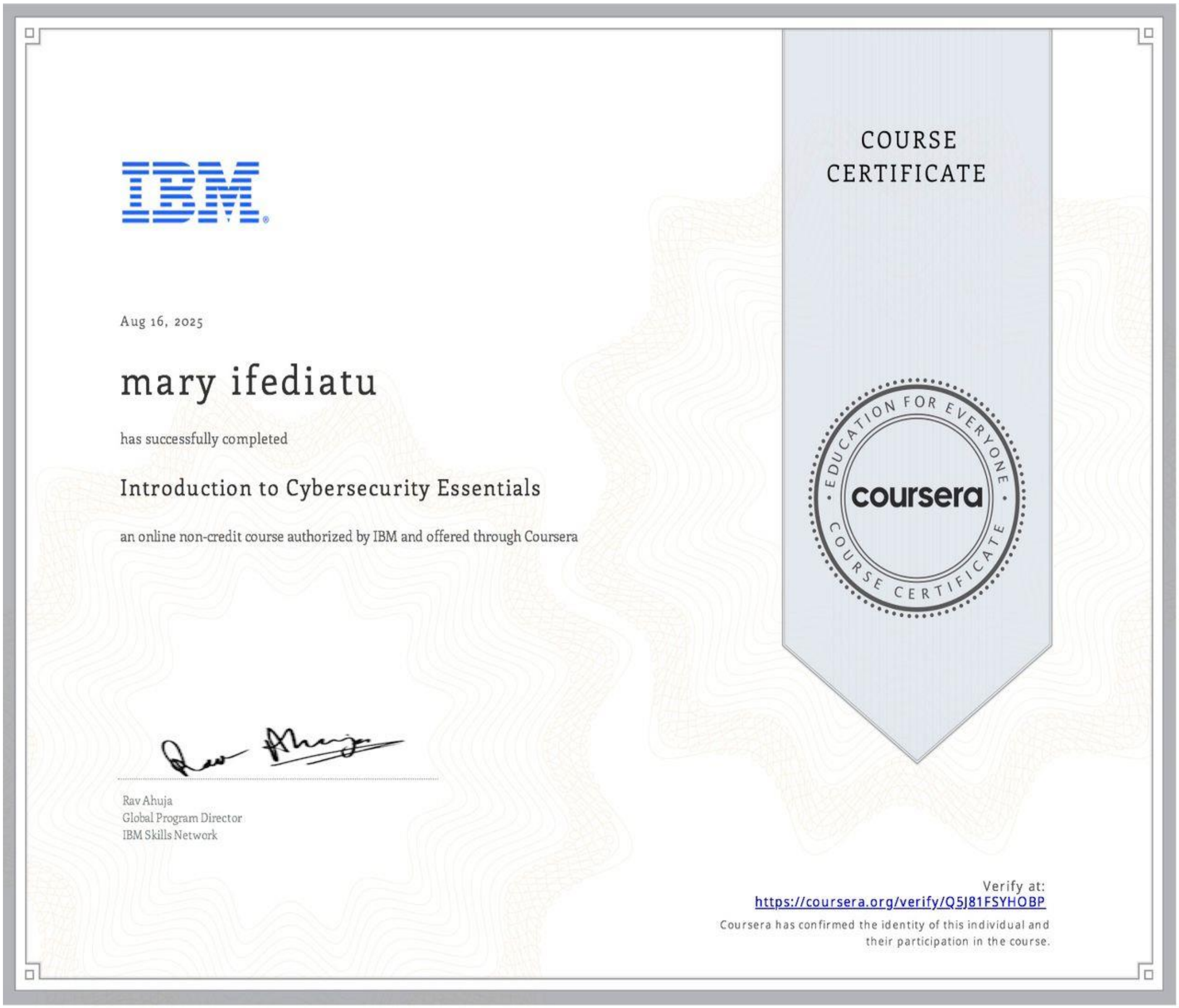
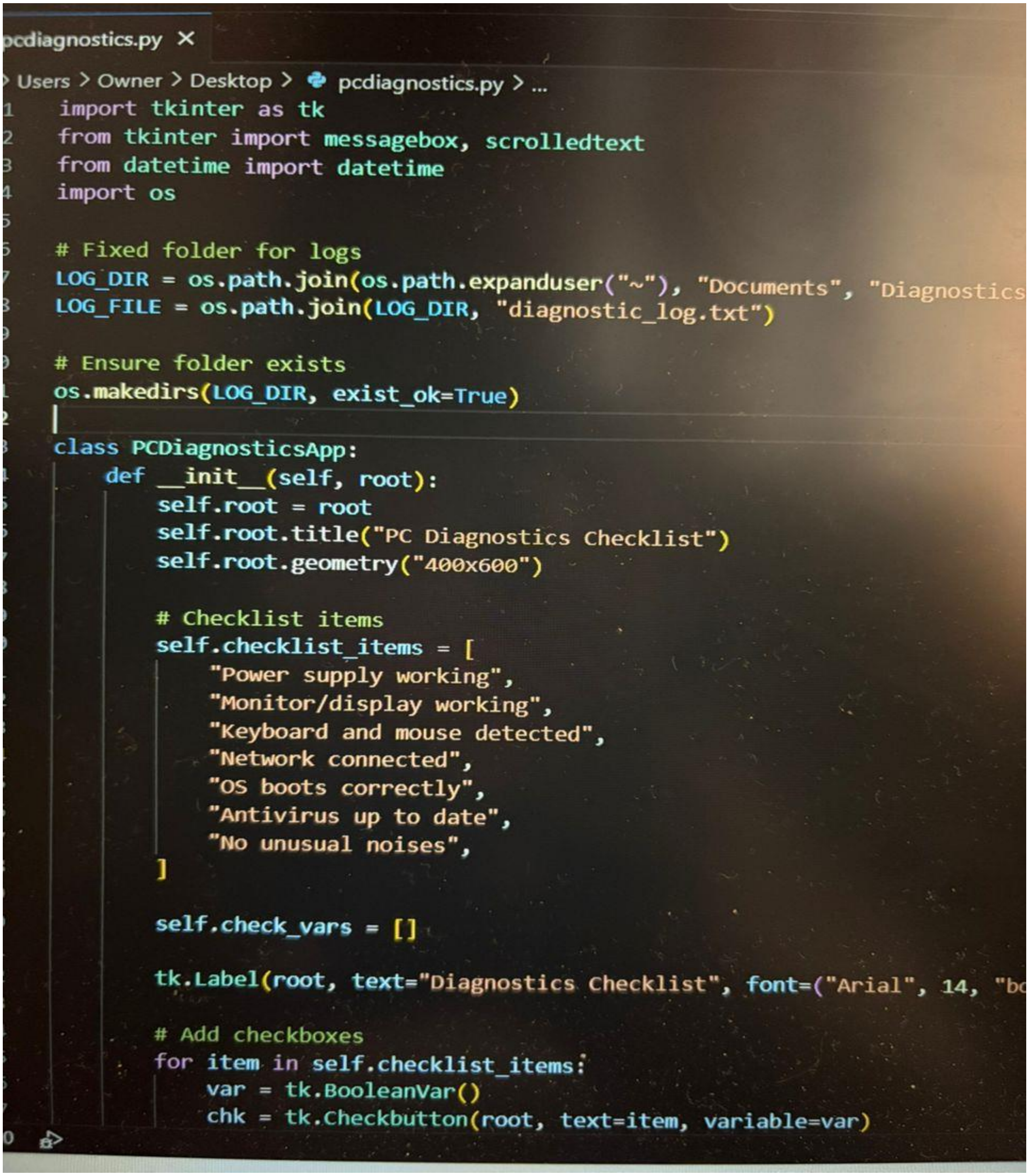


Figure 6.1: Certificate of Completion for the 10-Day "Introduction to Cybersecurity Essentials" Course.



```
pcdiagnostics.py ×
> Users > Owner > Desktop > pcdiagnostics.py > ...
import tkinter as tk
from tkinter import messagebox, scrolledtext
from datetime import datetime
import os

# Fixed folder for logs
LOG_DIR = os.path.join(os.path.expanduser("~"), "Documents", "Diagnostics
LOG_FILE = os.path.join(LOG_DIR, "diagnostic_log.txt")

# Ensure folder exists
os.makedirs(LOG_DIR, exist_ok=True)

class PCDiagnosticsApp:
    def __init__(self, root):
        self.root = root
        self.root.title("PC Diagnostics Checklist")
        self.root.geometry("400x600")

        # Checklist items
        self.checklist_items = [
            "Power supply working",
            "Monitor/display working",
            "Keyboard and mouse detected",
            "Network connected",
            "OS boots correctly",
            "Antivirus up to date",
            "No unusual noises",
        ]

        self.check_vars = []

        tk.Label(root, text="Diagnostics Checklist", font=("Arial", 14, "bo

        # Add checkboxes
        for item in self.checklist_items:
            var = tk.BooleanVar()
            chk = tk.Checkbutton(root, text=item, variable=var)
```

Figure 4.1: Python Code for the PC Diagnostics Checklist Tool developed using Tkinter.

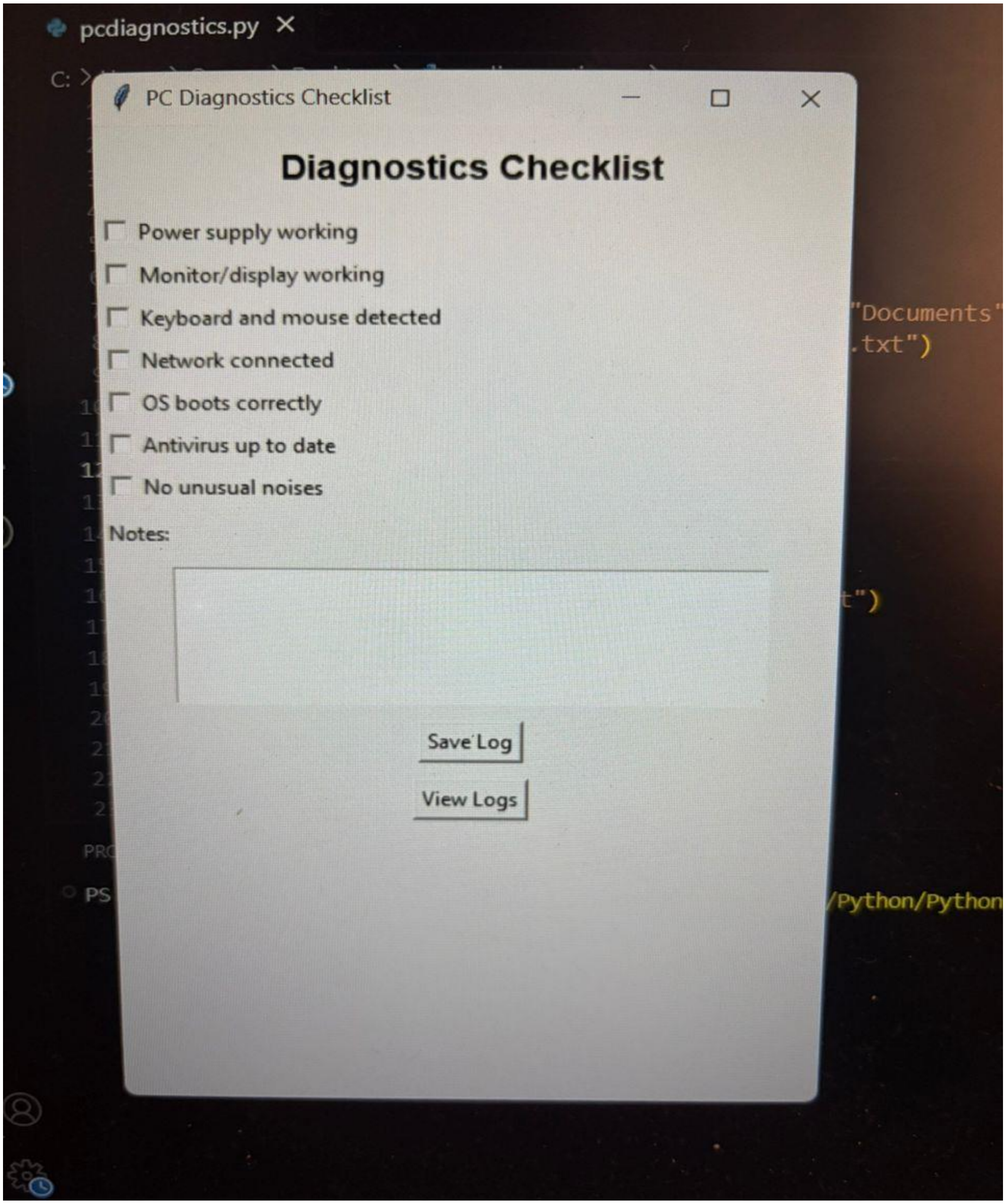


Figure 4.2: PC Diagnostics Checklist Tool Interface displaying hardware and software check options

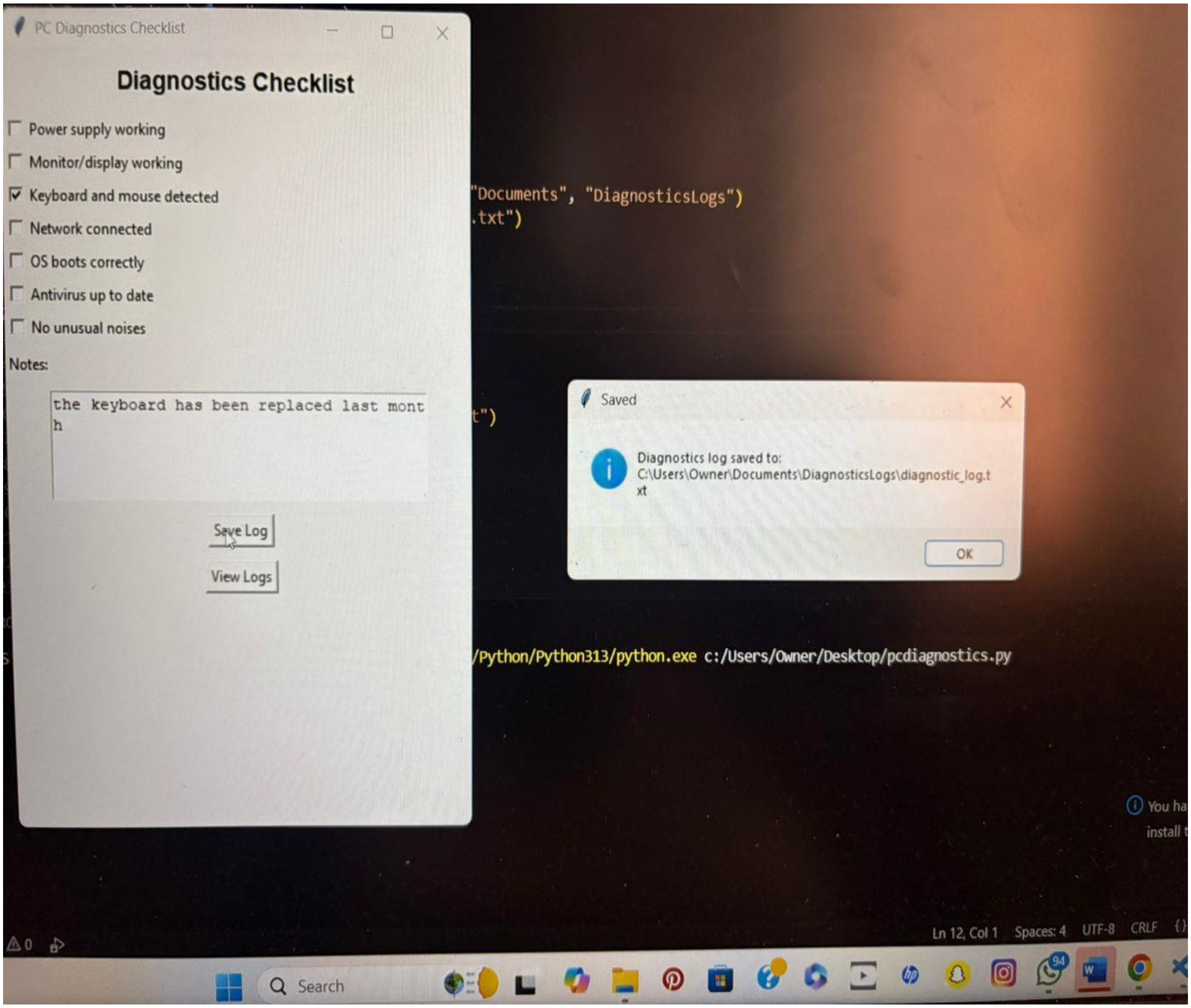


Figure 4.3: PC Diagnostics Checklist Tool Interface displaying where it is saved when clicked.

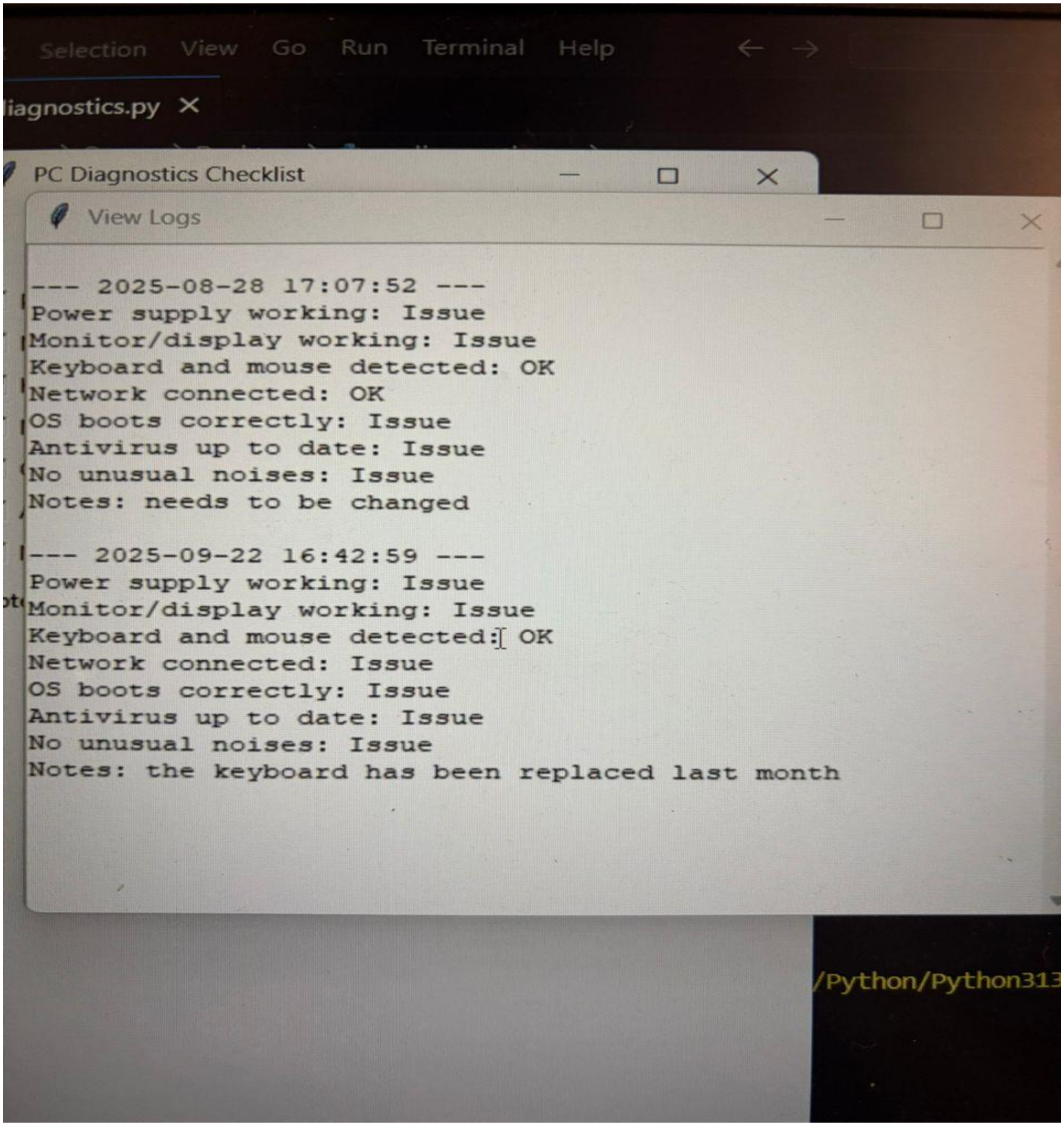
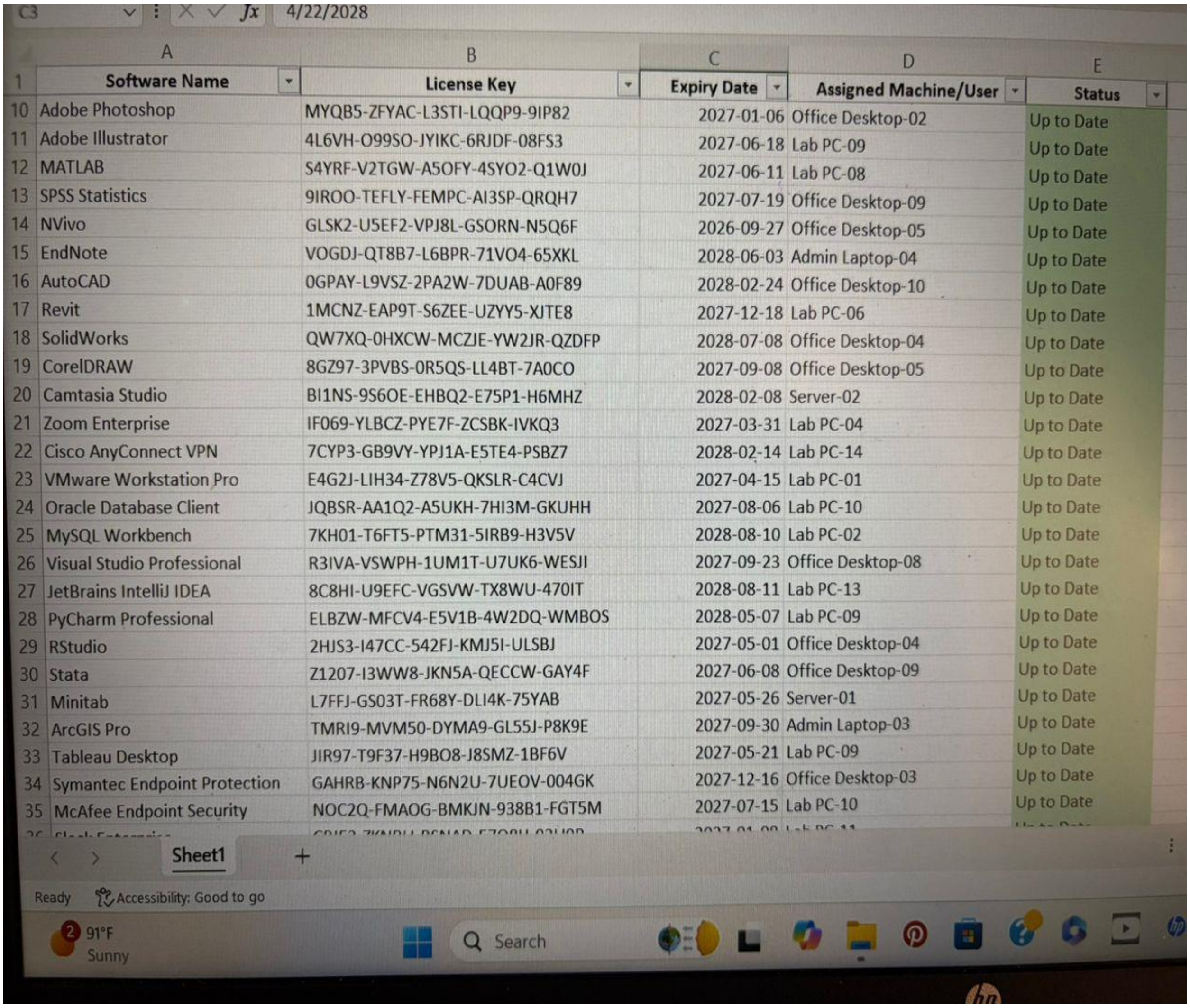


Figure 4.3: Diagnostic Log File generated automatically after saving system check results.



	Software Name	License Key	Expiry Date	Assigned Machine/User	Status
10	Adobe Photoshop	MYQB5-ZFYAC-L3STI-LQQP9-9IP82	2027-01-06	Office Desktop-02	Up to Date
11	Adobe Illustrator	4L6VH-O99SO-JYIKC-6RJDF-08FS3	2027-06-18	Lab PC-09	Up to Date
12	MATLAB	S4YRF-V2TGW-A5OFY-4SYO2-Q1W0J	2027-06-11	Lab PC-08	Up to Date
13	SPSS Statistics	9IROO-TEFLY-FEMPC-AI3SP-QRQH7	2027-07-19	Office Desktop-09	Up to Date
14	NVivo	GLSK2-U5EF2-VPJ8L-GSORN-N5Q6F	2026-09-27	Office Desktop-05	Up to Date
15	EndNote	VOGDJ-QT8B7-L6BPR-71VO4-65XKL	2028-06-03	Admin Laptop-04	Up to Date
16	AutoCAD	0GPAY-L9VSZ-2PA2W-7DUAB-A0F89	2028-02-24	Office Desktop-10	Up to Date
17	Revit	1MCNZ-EAP9T-S6ZEE-UZYY5-XJTE8	2027-12-18	Lab PC-06	Up to Date
18	SolidWorks	QW7XQ-0HXCW-MCZJE-YW2JR-QZDFP	2028-07-08	Office Desktop-04	Up to Date
19	CorelDRAW	8GZ97-3PVBS-0R5QS-LL4BT-7A0CO	2027-09-08	Office Desktop-05	Up to Date
20	Camtasia Studio	BI1NS-9S6OE-EHBQ2-E75P1-H6MHZ	2028-02-08	Server-02	Up to Date
21	Zoom Enterprise	IF069-YLBCZ-PYE7F-ZCSBK-IVKQ3	2027-03-31	Lab PC-04	Up to Date
22	Cisco AnyConnect VPN	7CYP3-GB9VY-YPJ1A-E5TE4-PSBZ7	2028-02-14	Lab PC-14	Up to Date
23	VMware Workstation Pro	E4G2J-LIH34-Z78V5-QKSLR-C4CVJ	2027-04-15	Lab PC-01	Up to Date
24	Oracle Database Client	JQBSR-AA1Q2-A5UKH-7HI3M-GKUHH	2027-08-06	Lab PC-10	Up to Date
25	MySQL Workbench	7KH01-T6FT5-PTM31-5IRB9-H3V5V	2028-08-10	Lab PC-02	Up to Date
26	Visual Studio Professional	R3IVA-VSWPH-1UM1T-U7UK6-WESJI	2027-09-23	Office Desktop-08	Up to Date
27	JetBrains IntelliJ IDEA	8C8HI-U9EFC-VGSVW-TX8WU-470IT	2028-08-11	Lab PC-13	Up to Date
28	PyCharm Professional	ELBZW-MFCV4-E5V1B-4W2DQ-WMBOS	2028-05-07	Lab PC-09	Up to Date
29	RStudio	2HJS3-I47CC-542FJ-KMJ5I-ULSBJ	2027-05-01	Office Desktop-04	Up to Date
30	Stata	Z1207-I3WW8-JKN5A-QECCW-GAY4F	2027-06-08	Office Desktop-09	Up to Date
31	Minitab	L7FFJ-GS03T-FR68Y-DLI4K-75YAB	2027-05-26	Server-01	Up to Date
32	ArcGIS Pro	TMRI9-MVM50-DYMA9-GL55J-P8K9E	2027-09-30	Admin Laptop-03	Up to Date
33	Tableau Desktop	JIR97-T9F37-H9BO8-J8SMZ-1BF6V	2027-05-21	Lab PC-09	Up to Date
34	Symantec Endpoint Protection	GAHRB-KNP75-N6N2U-7UEOV-004GK	2027-12-16	Office Desktop-03	Up to Date
35	McAfee Endpoint Security	NOC2Q-FMAOG-BMKJN-938B1-FGT5M	2027-07-15	Lab PC-10	Up to Date

Figure 3.1: Software License Inventory Spreadsheet showing software records and license details.

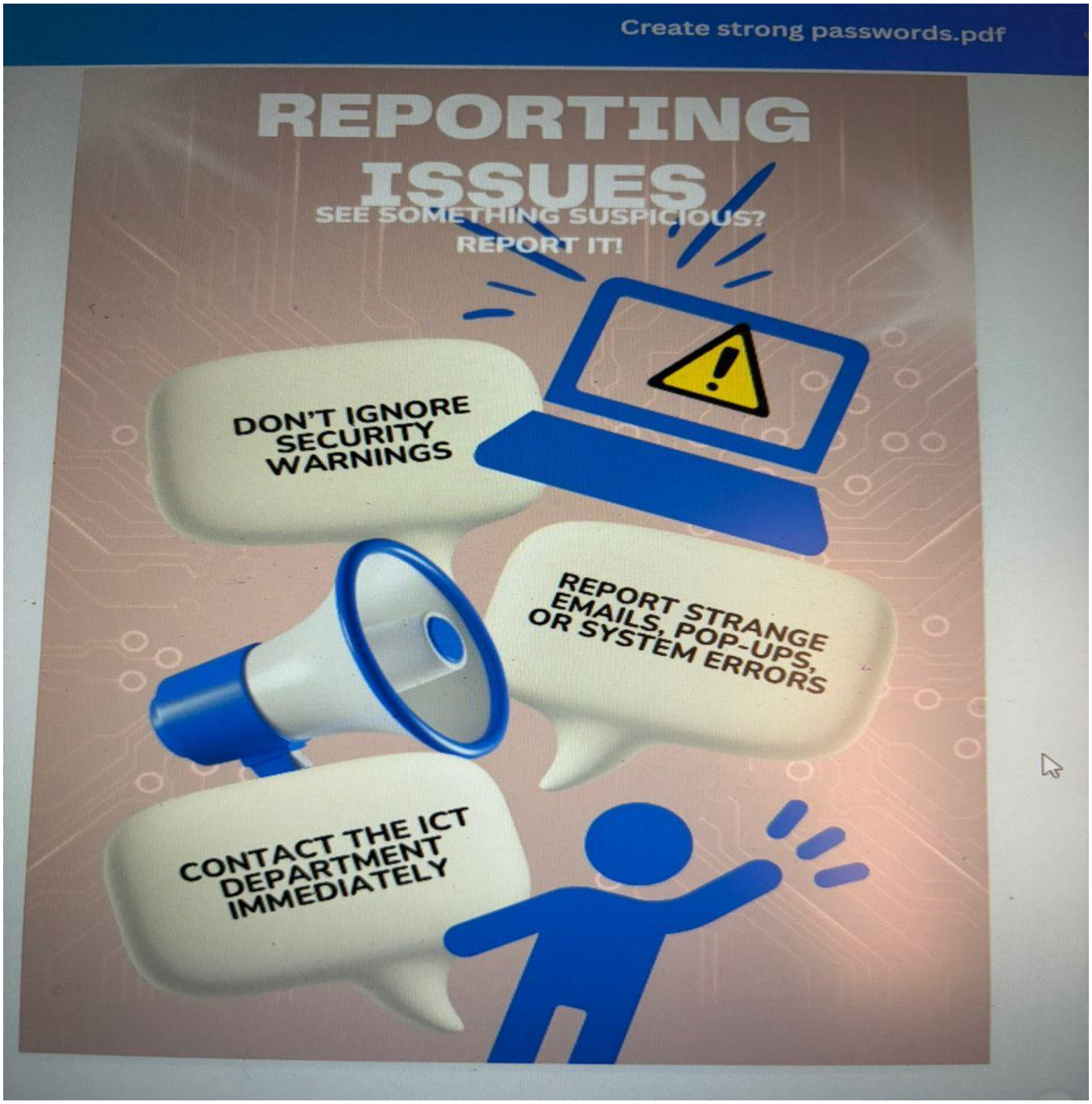


Figure 7.3: Cybersecurity Awareness Poster – *"See Something Suspicious? Report It!"*

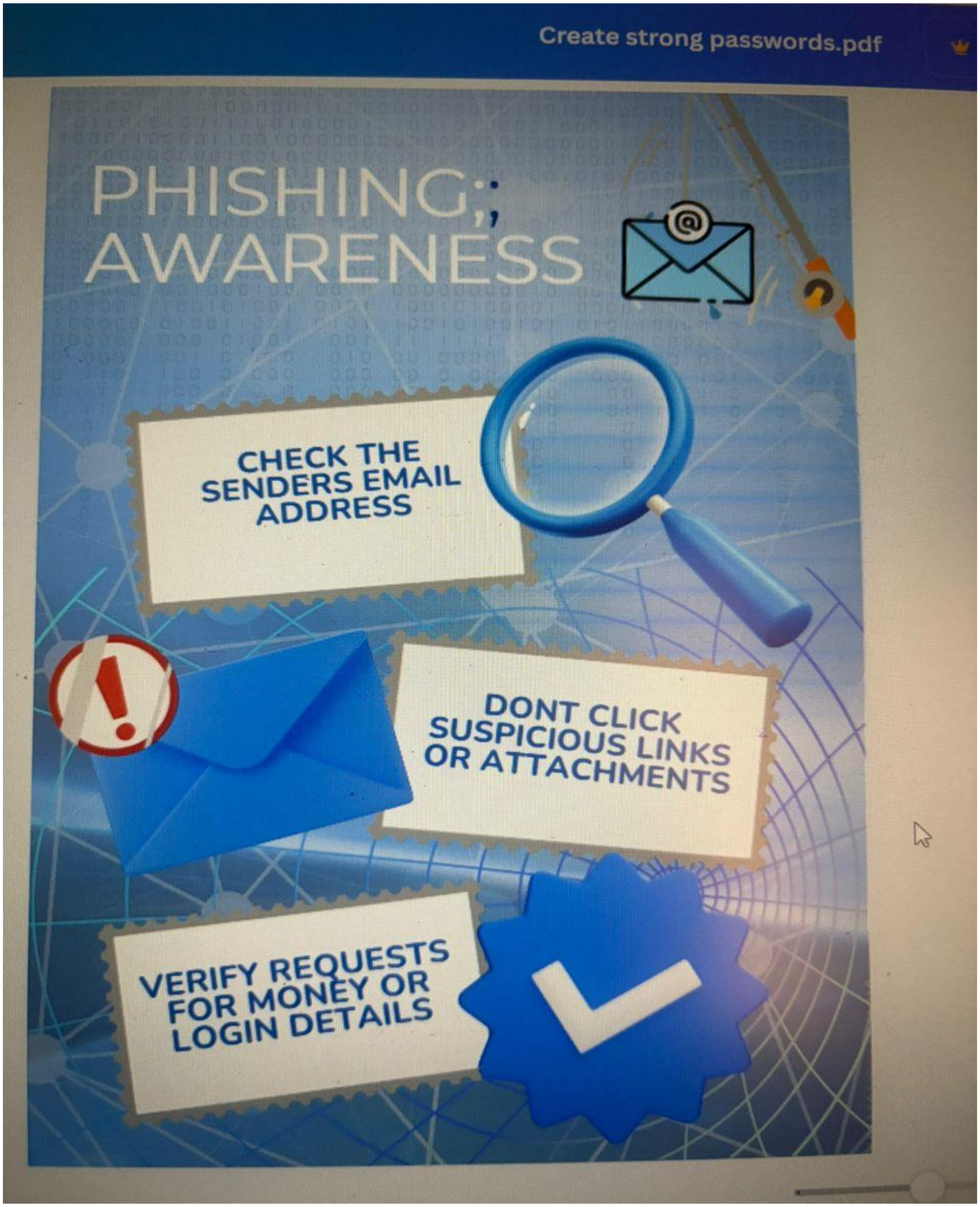


Figure 7.1: Cybersecurity Awareness Poster – ***"Don't Take the Bait: Spot Phishing Emails."***

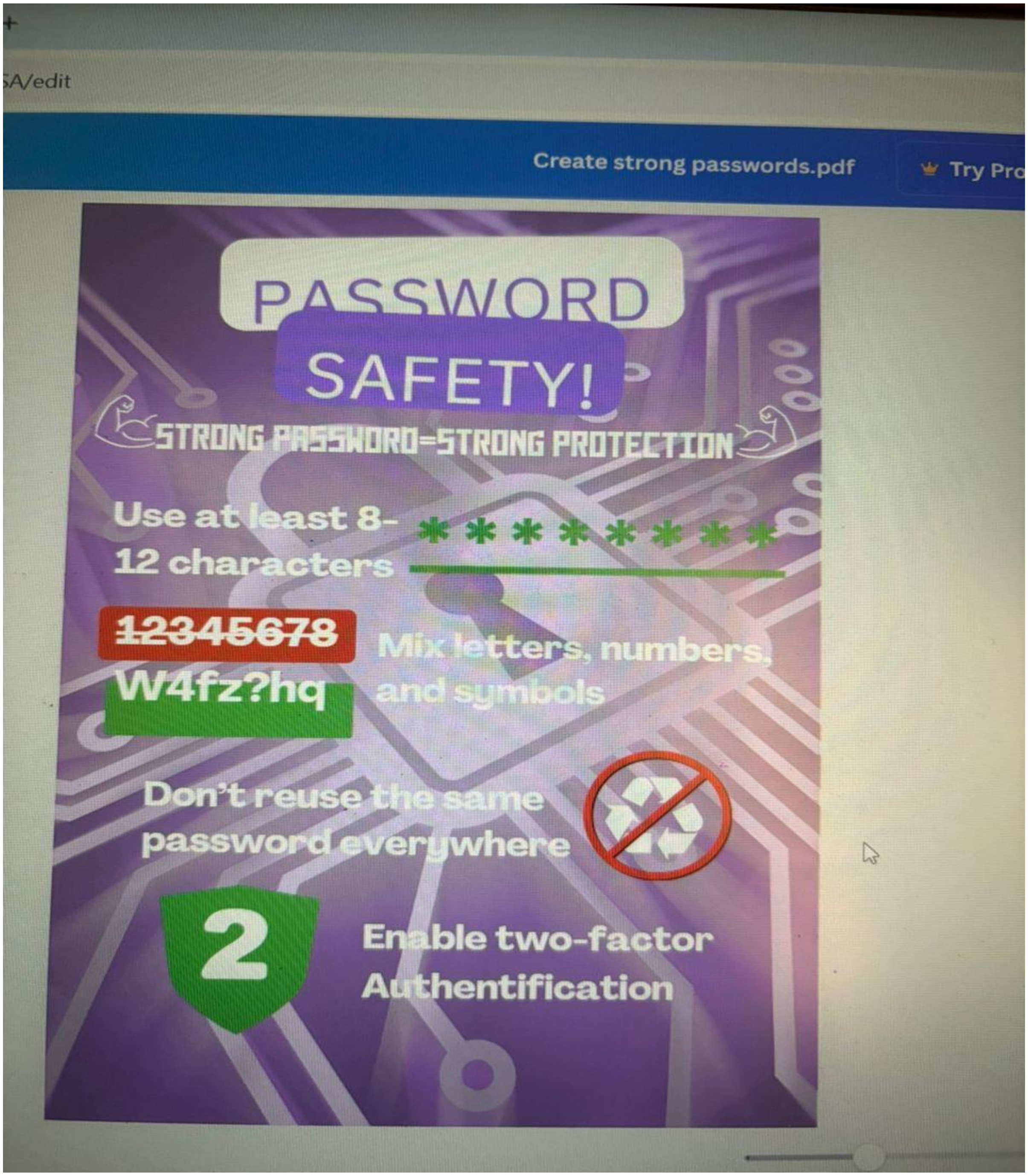


Figure 7.2: Cybersecurity Awareness Poster – ***"Strong Passwords = Strong Protection."***